

T3 group factors: UV

Active theory: SM + $S_1 + S_2 + F$. The same beta function is shown across runs in a single comparison table, with exact analytic group factors in the run columns.

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Field representations

run	S_1	S_2	F	α
T3 – B	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(1, 0)$	-1
T3 – C	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(3, 0)$	-1
T3 – A	$(1, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	0
T3 – D	$(3, -1)$	$(1, 0)$	$(2, -\frac{1}{2})$	-2
T3 – E	$(3, 0)$	$(3, 1)$	$(2, \frac{1}{2})$	0

β_{y_1}

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$\mathcal{G}_{\text{tr}}$	1	$\frac{3}{2}$	2	1	1
$\mathcal{G}_{\text{self}}$	$\frac{3}{2}$	$\frac{5}{4}$	1	$\frac{3}{2}$	$\frac{3}{2}$
$\mathcal{G}_{\text{cross}}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{3}{4}$
$\mathcal{G}_{Y_e}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$\mathcal{G}_{SU(2)_L}$	$-\frac{9}{4}$	$-\frac{33}{4}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$\mathcal{G}_{U(1)_Y}$	$-\frac{3}{4}$	$-\frac{3}{4}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$

β_{y_2}

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$\mathcal{G}_{\text{tr}}$	1	$\frac{3}{2}$	1	2	1
$\mathcal{G}_{\text{self}}$	$\frac{3}{2}$	$\frac{5}{4}$	$\frac{3}{2}$	1	$\frac{3}{2}$
$\mathcal{G}_{\text{cross}}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$
$\mathcal{G}_{Y_e}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$\mathcal{G}_{SU(2)_L}$	$-\frac{9}{4}$	$-\frac{33}{4}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$\mathcal{G}_{U(1)_Y}$	$-\frac{3}{4}$	$-\frac{3}{4}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$

β_{M_F}

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$(y_1^T y_1^*) M_F$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$
$M_F (y_2^\dagger y_2)$	1	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{3}{4}$
$g_2^2 M_F$	0	-12	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_Y^2 M_F$	0	0	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$

$\beta_{m_{S_1}^2}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
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$m_{S_1}^2 \text{Tr}(y_1^\dagger y_1)$	2	3	4	2	2
$M_F^\dagger M_F \text{Tr}(y_1^\dagger y_1)$	-16	-24	-8	-4	-4
$\lambda_{S_1} m_{S_1}^2$	6	6	4	8	8
$g_2^2 m_{S_1}^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	0	-12	-12
$g_Y^2 m_{S_1}^2$	$-\frac{3}{2}$	$-\frac{3}{2}$	0	-6	0
$\lambda_{12} m_{S_2}^2$	4	4	6	2	6
$\lambda_{S_1}^{\text{adj}} m_{S_1}^2$	0	0	0	4	4
$\lambda_{12}^{\text{cross}} m_{S_2}^2$	0	0	0	0	4

$\beta_{m_{S_2}^2}$

$\mathcal{S}$	T3 - B ($\alpha = -1$)	T3 - C ($\alpha = -1$)	T3 - A ($\alpha = 0$)	T3 - D ($\alpha = -2$)	T3 - E ($\alpha = 0$)
$m_{S_2}^2 \text{Tr}(y_2^\dagger y_2)$	2	3	2	4	2
$M_F^\dagger M_F \text{Tr}(y_2^\dagger y_2)$	-16	-24	-4	-8	-4
$\lambda_{S_2} m_{S_2}^2$	6	6	8	4	8
$g_2^2 m_{S_2}^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	-12	0	-12
$g_Y^2 m_{S_2}^2$	$-\frac{3}{2}$	$-\frac{3}{2}$	-6	0	-6
$\lambda_{12} m_{S_1}^2$	4	4	2	6	6
$\lambda_{S_2}^{\text{adj}} m_{S_2}^2$	0	0	4	0	4
$\lambda_{12}^{\text{cross}} m_{S_1}^2$	0	0	0	0	4

$\beta_{\lambda_{H1}}$

$\mathcal{S}$	T3 - B ($\alpha = -1$)	T3 - C ($\alpha = -1$)	T3 - A ($\alpha = 0$)	T3 - D ($\alpha = -2$)	T3 - E ($\alpha = 0$)
$\lambda_{H1} \text{Tr}(y_1^\dagger y_1)$	2	3	4	2	2
$\lambda_{H1} \text{Tr}(Y_d^\dagger Y_d)$	6	6	6	6	6
$\lambda_{H1} \text{Tr}(Y_e^\dagger Y_e)$	2	2	2	2	2
$\lambda_{H1} \text{Tr}(Y_u^\dagger Y_u)$	6	6	6	6	6
$\text{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$	-2	-3	-4	-2	-2
$ \lambda_{T3} ^2$	6	6	12	4	4
$g_2^2 \lambda_{H1}$	-9	-9	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_Y^2 \lambda_{H1}$	-3	-3	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$
$\lambda_H \lambda_{H1}$	6	6	6	6	6
$(\lambda_{H1})^2$	4	4	4	4	4
$\lambda_{H1} \lambda_{S1}$	6	6	4	8	8
$\lambda_{12} \lambda_{H2}$	4	4	6	2	6
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	0	6	6
g_Y^4	$\frac{3}{4}$	$\frac{3}{4}$	0	3	0
$(\lambda_{H1}^{\text{adj}})^2$	$\frac{3}{4}$	$\frac{3}{4}$	0	2	2
$\lambda_{H1} \lambda_{S1}^{\text{adj}}$	0	0	0	4	4
$\lambda_{H2} \lambda_{12}^{\text{cross}}$	0	0	0	0	4

$\beta_{\lambda_{H2}}$

$\mathcal{S}$	T3 - B ($\alpha = -1$)	T3 - C ($\alpha = -1$)	T3 - A ($\alpha = 0$)	T3 - D ($\alpha = -2$)	T3 - E ($\alpha = 0$)
$\lambda_{H2} \text{Tr}(y_2^\dagger y_2)$	2	3	2	4	2
$\lambda_{H2} \text{Tr}(Y_d^\dagger Y_d)$	6	6	6	6	6
$\lambda_{H2} \text{Tr}(Y_e^\dagger Y_e)$	2	2	2	2	2
$\lambda_{H2} \text{Tr}(Y_u^\dagger Y_u)$	6	6	6	6	6
$\text{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$	-2	-3	-2	-4	-2
$ \lambda_{T3} ^2$	6	6	4	12	4

$g_2^2 \lambda_{H2}$	-9	-9	$-\frac{33}{2}$	$-\frac{9}{2}$	$-\frac{33}{2}$
$g_Y^2 \lambda_{H2}$	-3	-3	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$
$\lambda_H \lambda_{H2}$	6	6	6	6	6
$(\lambda_{H2})^2$	4	4	4	4	4
$\lambda_{H2} \lambda_{S2}$	6	6	8	4	8
$\lambda_{12} \lambda_{H1}$	4	4	2	6	6
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	6	0	6
g_Y^4	$\frac{3}{4}$	$\frac{3}{4}$	3	0	3
$\left(\lambda_{H2}^{\text{adj}}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	2	0	2
$\lambda_{H2} \lambda_{S2}^{\text{adj}}$	0	0	4	0	4
$\lambda_{H1} \lambda_{12}^{\text{cross}}$	0	0	0	0	4

$\beta_{\lambda_{12}}$

$\mathcal{S}$	T3 - B ($\alpha = -1$)	T3 - C ($\alpha = -1$)	T3 - A ($\alpha = 0$)	T3 - D ($\alpha = -2$)	T3 - E ($\alpha = 0$)
$\lambda_{12} \text{Tr}(y_1^\dagger y_1)$	2	3	4	2	2
$\lambda_{12} \text{Tr}(y_2^\dagger y_2)$	2	3	2	4	2
$\text{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$	-2	$-\frac{9}{2}$	-4	-4	-2
$ \lambda_{T3} ^2$	3	3	4	4	$\frac{4}{3}$
$g_2^2 \lambda_{12}$	-9	-9	-12	-12	-24
$g_Y^2 \lambda_{12}$	-3	-3	-6	-6	-6
$(\lambda_{12})^2$	4	4	4	4	4
$\lambda_{12} \lambda_{S1}$	6	6	4	8	8
$\lambda_{12} \lambda_{S2}$	6	6	8	4	8
$\lambda_{H1} \lambda_{H2}$	4	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	0	0	12
g_Y^4	$\frac{3}{4}$	$\frac{3}{4}$	0	0	0
$\left(\lambda_{12}^{\text{adj}}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	0	0	4
$\lambda_{12} \lambda_{S2}^{\text{adj}}$	0	0	4	0	4
$\lambda_{12} \lambda_{S1}^{\text{adj}}$	0	0	0	4	4
$(\lambda_{12}^{\text{cross}})^2$	0	0	0	0	4
$\lambda_{12}^{\text{cross}} \lambda_{S1}^{\text{adj}}$	0	0	0	0	4
$\lambda_{12}^{\text{cross}} \lambda_{S2}^{\text{adj}}$	0	0	0	0	4

$\beta_{\lambda_{T3}}$

$\mathcal{S}$	T3 - B ($\alpha = -1$)	T3 - C ($\alpha = -1$)	T3 - A ($\alpha = 0$)	T3 - D ($\alpha = -2$)	T3 - E ($\alpha = 0$)
$\lambda_{T3} \text{Tr}(Y_d^\dagger Y_d)$	6	6	6	6	6
$\lambda_{T3} \text{Tr}(Y_e^\dagger Y_e)$	2	2	2	2	2
$\lambda_{T3} \text{Tr}(Y_u^\dagger Y_u)$	6	6	6	6	6
$\lambda_{T3} \lambda_H$	2	2	2	2	2
$\lambda_{T3} \lambda_{H1}$	4	4	4	4	4
$\lambda_{T3} \lambda_{H2}$	4	4	4	4	4
$\lambda_{T3} \lambda_{12}$	2	2	2	2	2
$g_2^2 \lambda_{T3}$	-9	-9	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{33}{2}$
$g_Y^2 \lambda_{T3}$	-3	-3	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$\lambda_{T3} \text{Tr}(y_1^\dagger y_1)$	1	$\frac{3}{2}$	2	1	1
$\lambda_{T3} \text{Tr}(y_2^\dagger y_2)$	1	$\frac{3}{2}$	1	2	1
$\lambda_{T3} \lambda_{H1}^{\text{adj}}$	-2	-2	0	-4	-2
$\lambda_{T3} \lambda_{H2}^{\text{adj}}$	2	2	4	0	2
$\lambda_{T3} \lambda_{12}^{\text{adj}}$	$-\frac{1}{2}$	$-\frac{1}{2}$	0	0	2
$\lambda_{T3} \lambda_{12}^{\text{cross}}$	0	0	0	0	-2

generated non-singlet sectors

S	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$\beta_{\lambda_{H1}^{\text{adj}}} : \text{g2sq_gYsq}$	−6	−6	0	−12	0
$\beta_{\lambda_{H2}^{\text{adj}}} : \text{g2sq_gYsq}$	6	6	12	0	12
$\beta_{\lambda_{12}^{\text{adj}}} : \text{g2sq_gYsq}$	−6	−6	0	0	0
$\beta_{\lambda_{12}^{\text{cross}}} : g_2^4$	0	0	0	0	6
$\beta_{\lambda_{12}^{\text{cross}}} : \left(\lambda_{12}^{\text{adj}}\right)^2$	0	0	0	0	2
$\beta_{\lambda_{12}^{\text{cross}}} : \lambda_{12} \lambda_{12}^{\text{cross}}$	0	0	0	0	8
$\beta_{\lambda_{12}^{\text{cross}}} : \left(\lambda_{12}^{\text{cross}}\right)^2$	0	0	0	0	10
$\beta_{\lambda_{12}^{\text{cross}}} : \lambda_{12}^{\text{cross}} \lambda_{S1}^{\text{adj}}$	0	0	0	0	−2
$\beta_{\lambda_{12}^{\text{cross}}} : \lambda_{12}^{\text{cross}} \lambda_{S2}^{\text{adj}}$	0	0	0	0	−2

Threshold interpretation

At $\mu = M_F$, the fermion F is integrated out. Therefore y_1 , y_2 and M_F cease to be active EFT parameters and their information is transferred to the matched $C_{LLS_1S_2}$ boundary tensor.

Convention note

All numerical group factors above are read from the analytic group-factor modules used by the pipeline; the report does not hard-code A–E coefficients.